8 bit → rappr. in complemento a 2

$32 - 36 = -4$

A = 32 ⟶ 0010 0000

B = -36 36 → binario $36 = 32 + 4 = (0010\ 0100)_2$

0010 0100 NOT
1101 1011 +
1 =

B = 1101 1100 = −36

A = 32, 0010 0000 +
B = -36 1101 1100 =

(1)1111 1100 < 0
≡ −4

0000 0011 +
1
0000 0100 = 4

$36 - 32 = 4$

$36 = 32 + 4 = (0010\ 0100)_2$

$32 \rightarrow$ 0010 0000 NOT

1101 1111 +

1

11100000 $\equiv -32$

(1 1 1

36 0010 0100 +

-32 11100000 =

100000100 $= 2^2 = 4$

$$13,65 = 1\cdot 10^{1} + 3\cdot 10^{0} + 6\cdot 10^{-1} + 5\cdot 10^{-2}$$

$$(1011,101)_2 = 1\cdot 2^{3} + 1\cdot 2^{1} + 1\cdot 2^{0} + 1\cdot 2^{-1} + 1\cdot 2^{-3} =$$

$$= \underbrace{2^3 + 2 + 1} + \underbrace{\frac{1}{2} + \frac{1}{2^3}} =$$

$$= 11 + 0.5 + \frac{1}{8} = 11 + 0.5 + 0.125 =$$

$$= 11,625$$

— o —

$11,625$ → binario

$11 = (1011)_2$

| | 2 |
|---|---|
| 11 | 1 |
| 5 | 1 |
| 2 | 0 |
| 1 | 1 |
| 0 | |

$\times 2$

| | |
|---|---|
| 0,625 | 1 |
| 0,250 | 0 |
| 0,5 | 1 |
| 0 | |

$0,625 \times 2 = 1,25$

$0,25 \times 2 = 0,5$

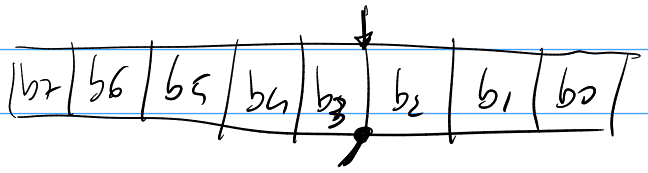
$0,5 \times 2 = 1$

$$0,625 = (0,101)_2$$

$$11,625 = (1011,101)_2$$

# VIRGOLA FISSA su N-bit

8 bit, VIRGOLA b2 + 3




10111,001 → VIRGOLA FISSA
N dec. = 3

$(10111)_2 = 2^4 + 2^2 + 2^1 + 2^0 =$
$= 16 + 4 + 2 + 1 = 23$

$(,001)_2 = 2^{-3} = \frac{1}{2^3} = \frac{1}{8} = 0,125$

$(10111001)_2 = 23,125$

$1,5 = (1,1)_2$ $\quad 2^0 + 2^{-1} = 1 + \frac{1}{2} =$
$= 1 + 0,5 =$
$= 1,5$

$23,125 + 1,5 = 24,625$

$10111001 + 00001100 = 11000101 = 24,625$

$2^3 + 2^4 \quad 2^{-1} + 2^{-3}$

$16 + 8 = 24 \quad \frac{1}{2} + \frac{1}{8} = 0,5 + 0,125 =$
$= 0,625$

b7 b6 b5 b4 b3 b2 b1 b0

# VIRGOLA MOBILE

$0,00000013 \rightarrow 13 \cdot 10^{-8}$

$13000000000 = 13 \cdot 10^{10}$

$6 \cdot 10^{-5} + 8 \cdot 10^{-4} = 86 \cdot 10^{-5}$

0,00006 0,0008

0,00006
0,0008
0,00086

$6 \cdot 10^{-5} + 8 \cdot 10^{-4} = 6 \cdot 10^{-1} \cdot 10^{-4} + 8 \cdot 10^{-4} =$

$= (6 \cdot 10^{-1} + 8) \cdot 10^{-4} =$

$= (0,6 + 8) \cdot 10^{-4} = 8,6 \cdot 10^{-4} =$

$= 86 \cdot 10^{-5}$

$\times 2$

| | | |
|---|---|---|
| | $0,3$ | $0$ |
| | $0,6$ | $1$ |
| $1,2$ | $0,2$ | $0$ |
| | $0,4$ | $0$ |
| | $0,8$ | $0$ |
| $1,6$ | $0,6$ | $1$ |
| | | $0$ |
| | | $\vdots$ |

$0,3 = (0,\overline{1000})_2$